

Modelling the noise influence associated with the discrete linear canonical transform

ISSN 1751-9675

Received on 8th July 2017

Revised 24th December 2017

Accepted on 25th February 2018

E-First on 4th May 2018

doi: 10.1049/iet-spr.2017.0319

www.ietdl.org

Yi-Ping Bao¹, Bing-Zhao Li¹ ✉¹School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 102488, People's Republic of China

✉ E-mail: li_bingzhao@bit.edu.cn

Abstract: In this study, the properties of noise after a discrete linear canonical transform (DLCT) are analysed. First, the authors prove that the DLCT of noise can be modelled by a Gaussian distribution with very weak assumptions on the noise in the time domain. Then, the mean and covariance matrix of this Gaussian distribution is derived and the general trend of the noise after DLCT is described. In addition, they find that the properties of noise in the LCT domain are the generalisation of the properties of noise in the Fourier transform domain. What is more, the authors prove that the additive white Gaussian noise (AWGN) is still an AWGN after performing the DLCT. Finally, the simulations are performed to verify the effectiveness of the obtained results.

1 Introduction

The linear canonical transform (LCT) is a three-parameter linear integral transform [1–4]. It was introduced in the 1970s and many transforms, such as the Fourier transform (FT), the fractional FT (FRFT) and the Fresnel transform, are special cases of the LCT [3, 5–7]. The continuous LCT has been used in a number of applications, e.g. in optical systems, filter design, graded-index media analysis, time–frequency analysis and holographic three-dimensional television [8–14]. As LCT is widely used in many subjects, many kinds of definitions and implementation of the discrete LCT (DLCT) have been derived in [15–21].

To use the results of these algorithms in an efficient way, it is necessary to know the value of signal in the LCT domain due to the noise disturbance. The noise influence after a discrete Fourier transform (DFT) has been discussed before [22–24], and the theory has been used in electric power system [25], frequency estimation [26], optics [27] and so on. However, to the best of our knowledge, no paper has been published associated with the noise influence in the LCT domain. It is therefore interesting and worthwhile to investigate the noise influence after DLCT.

In this paper, we demonstrate that the noise on the LCT domain can be approximately modelled as Gaussian distribution. The mean and covariance matrix, which determine the Gaussian distribution completely, can also be calculated. After the calculation, the general trend of the noise after DLCT is described. The properties of noise in the LCT domain can be seen as the generalisation of the properties of noise in the FT domain. In addition, the additive white Gaussian noise (AWGN) is still an AWGN after performing the DLCT. It is an important property and will be widely used in further research.

The organisation of this paper is as follows. Section 2 shows the definition of LCT and DLCT used in this passage. In Section 3, we make some assumptions on noise and prove the noise on the LCT domain can be approximately modelled as Gaussian in theory, and then we calculate the mean and covariance matrix of noise. Section 4 introduces some special cases. In Section 5, the proven theories are verified through computer simulations. Section 6 talks about some potential applications. The summary is provided in Section 7.

2 Preliminaries

2.1 Linear canonical transform

Definition 1: The LCT of a signal $f(t)$ with parameter matrix A is defined as

$$L_f^A(u) = L_A[f(t)](u) = \begin{cases} \int_{\mathbb{R}} f(t) K_A(u, t) dt, & b \neq 0 \\ \sqrt{d} \exp\left(\frac{icdu^2}{2}\right) f(du), & b = 0 \end{cases} \quad (1)$$

with

$$K_A(u, t) = \frac{1}{\sqrt{i2\pi b}} \exp\left(i \frac{at^2 - 2ut + du^2}{2b}\right),$$

where $A = (ab; cd)$ is the parameter matrix, a, b, c, d are real numbers and $\det(A) = ad - bc = 1$.

Note that when $b = 0$, the LCT becomes a chirp multiplication. Therefore, we only consider the case of $b \neq 0$. When $(a, b; c, d) = (\cos \alpha \sin \alpha; -\sin \alpha \cos \alpha)$, the LCT reduces to the FRFT and when $(a, b; c, d) = (0, 1; -1, 0)$, the LCT reduces to FT. For further details about properties of the LCT, readers can refer to [1, 2].

2.2 Discrete linear canonical transform

Nowadays, there are lots of DLCT methods. The method in [3] is the most basic one among them, and others are expansion of it [28]. So in this subsection we introduce a simple definition of the DLCT referred to [3].

Definition 2: Considering a signal $x(t)$, where $x(t) = 0$ for $T > 0$, $|t| > T$, we have

$$L_A(m) = \exp\left(\frac{i2dm^2b\pi^2}{\Delta t^2(2M+1)^2}\right) \times \sum_{n=-N}^N x(n) \exp\left(\frac{ian^2\Delta t^2}{2b} - \frac{i2\pi nm \operatorname{sgn}(b)}{2M+1}\right) \quad (2)$$

Meanwhile, the inverse transform is

$$x(n) = \frac{1}{2M+1} \exp\left(\frac{-ian^2\Delta t^2}{2b}\right) \sum_{m=-M}^M L_A(m) \times \exp\left(\frac{-i2dm^2b\pi^2}{\Delta t^2(2M+1)^2} - \frac{i2\pi nm \operatorname{sgn}(b)}{2M+1}\right) \quad (3)$$

where

$$\begin{aligned} x(n) &= x(n\Delta t), \quad n = -N, \dots, 0, 1, \dots, N; \\ L_A(m) &= L_A(m\Delta u), \quad m = -M, \dots, 0, 1, \dots, M; \end{aligned} \quad (4)$$

With $\Delta u = (2|b|\pi)/((2M+1)\Delta t)$. In this paper, in order to simplify the calculation, we only consider the case that $M = N$ and $b > 0$.

3 Noise influence on DLCT

3.1 Assumptions of noise

It is necessary to make some assumptions about the character of the time-domain noise. In order to obtain the general results, it is requisite to minimise the imposed restrictions. In this paper, the following assumptions on the noise in the time domain are made:

(1) We define that $y(*)$ is the measurement result; $x(*)$ is the true value and $n(*)$ is the noise contribution. So we have

$$y(k) = x(k) + n(k), \quad k \in [-N, N] \quad (5)$$

(2) The noise is stationary, which means that its statistical properties do not change in time. As a consequence, the autocorrelation of $n(k)$ is able to be expressed as

$$E[n(k)n(l)] = p_{nn}(k-l) \quad (6)$$

The mean of the noise $n(k)$ is

$$E[n(k)] = u_n \quad (7)$$

3.2 Distribution of noise after DLCT

Let a_m be the DLCT of the noise $n(k)$. In most DLCT methods [15–21], a_m can be written as the following form:

$$a_m = \sum_{k=-N}^N c(k), \quad m = -N, \dots, N \quad (8)$$

For example, using the method mentioned in Section 2.2, a_m is written as

$$\begin{aligned} a_m &= \exp\left(\frac{i2dm^2b\pi^2}{(2N+1)^2\Delta t^2}\right) \sum_{k=-N}^N n(k) \\ &\times \exp\left(\frac{iak^2\Delta t^2}{2b} - \frac{i2\pi km}{2N+1}\right), \quad m = -N, \dots, N \end{aligned} \quad (9)$$

and

$$\begin{aligned} c(k) &= n(k) \exp\left(\frac{iak^2\Delta t^2}{2b} + \frac{i2dm^2b\pi^2}{(2N+1)^2\Delta t^2}\right. \\ &\quad \left. - \frac{i2\pi km}{2N+1}\right), \quad k = -N, \dots, N \end{aligned} \quad (10)$$

The $n(k)$ are random variables, so the variables $c(k)$ are also random variables. From (8) it is seen that the noise a_m is come from a sum of a great number of random variables. The central limit theorem states: ‘a sum of a great number of random variables with a limited autocorrelation function is a new random variable with an asymptotic Gaussian distribution’. That is to say, the noise a_m satisfies a Gaussian distribution regardless of the distribution of the noise, when N is sufficiently high.

What is more, according to the theory, the theorems have nothing to do with the kind of the DLCT method, so in the following we will use the DLCT method mentioned in Section 2.2 to complete our calculation.

A Gaussian distribution is completely characterised by its mean and its covariance matrix. In next subsections, the mean and covariance matrix of noise will be calculated, and the noise can be described in theory by the results.

3.3 Mean

The mean of noise after DLCT shows the general trend of the noise in LCT domain. Therefore, in this subsection the mean of the noise after DLCT is studied. The expression of the mean of noise is given first. Then it will be shown that the noise after DLCT is similar to a certain chirp signal on FT domain which is only related to the parameter matrix A and the number of sampling points N . With the property of chirp signal and noise itself, the general trend of the noise after DLCT can be described.

For each $m \in [-N, N]$, we define $E[a_m]$ as the mean of a_m . Therefore, $E[a_m]$ is given below: (see (11)). This is the expression of the mean of noise after DLCT. Meanwhile, (11) can be expressed as

$$E[a_m] = \sum_{k=-N}^N f(k) \exp\left(-\frac{i2\pi km}{2N+1}\right) \quad (12)$$

where

$$\begin{aligned} f(t) &= u_n \exp\left(\frac{iat^2}{2b} + \frac{i2dm^2b\pi^2}{(2N+1)^2\Delta t^2}\right) \\ f(k) &= f(k\Delta t) \end{aligned} \quad (13)$$

Equation (12) is regarded as the DFT of $f(t)$. As we can see in (13), $f(t)$ is a chirp signal which initial frequency is 0 Hz and the frequency rate is $a/2b\pi$. So the general trend of the noise is similar to the chirp signal $f(t)$ on FT domain. According to the property of chirp signal [29, 30], the DLCT noise's components are zero if they are not in the interval $[0, a(2N+1)\Delta t/2b\pi]$. In other words, DFT

$$\begin{aligned} E[a_m] &= E\left[\exp\left(\frac{i2dm^2b\pi^2}{(2N+1)^2\Delta t^2}\right)\right. \\ &\quad \times \sum_{k=-N}^N n(k) \exp\left(\frac{iak^2\Delta t^2}{2b} - \frac{i2\pi km}{2N+1}\right)\Bigg] \\ &= \exp\left(\frac{i2dm^2b\pi^2}{(2N+1)^2\Delta t^2}\right) \\ &\quad \times E\left[\sum_{k=-N}^N n(k) \exp\left(\frac{iak^2\Delta t^2}{2b} - \frac{i2\pi km}{2N+1}\right)\right] \\ &= u_n \sum_{k=-N}^N \exp\left(\frac{iak^2\Delta t^2}{2b} - \frac{i2\pi km}{2N+1}\right. \\ &\quad \left. + \frac{i2dm^2b\pi^2}{(2N+1)^2\Delta t^2}\right), \quad m = -N, \dots, N \end{aligned} \quad (11)$$

coefficients of a signal are partly influenced by the noise if they are only in this interval. What is more, u_n can be easily derived from the knowledge of the noise properties in the time domain so finally $E[a_m]$ is derived by DFT method and the knowledge of chirp signal.

3.4 Covariance matrix

In this subsection, the covariance matrix of the noise after DLCT is studied. For each $m, r \in [-N, N]$, $\text{cov}(a_m, a_r) = E[(a_m - E[a_m])(a_r - E[a_r])] = E[a_m \bar{a}_r] - E[a_m]E[\bar{a}_r]$ ($\bar{a}_r$ is the conjugation of a_r), two terms can be distinguished. The first term $E[a_m]E[\bar{a}_r]$ is

$$\begin{aligned} E[a_m]E[\bar{a}_r] &= E\left[\exp\left(\frac{i2dm^2b\pi^2}{(2N+1)^2\Delta t^2}\right)\right. \\ &\quad \times \sum_{k=-N}^N n(k)\exp\left(\frac{iak^2\Delta t^2}{2b} - \frac{i2\pi km}{2N+1}\right) \\ &\quad \times E\left[\exp\left(-\frac{i2dr^2b\pi^2}{(2N+1)^2\Delta t^2}\right)\sum_{l=-N}^N n(l)\right. \\ &\quad \times \exp\left(-\frac{ial^2\Delta t^2}{2b} + \frac{i2\pi lr}{2N+1}\right) \\ &= u_n^2 \exp\left(\frac{i2d(m^2 - r^2)b\pi^2}{(2N+1)^2\Delta t^2}\right) \\ &\quad \times \sum_{k=-N}^N \sum_{l=-N}^N \exp\left(\frac{ia(k^2 - l^2)\Delta t^2}{2b} - \frac{i2\pi(km - lr)}{2N+1}\right) \end{aligned} \quad (14)$$

The second part $E[a_m \bar{a}_r]$ is the autocorrelation of a_m and a_r , it is transformed to

$$\begin{aligned} E[a_m \bar{a}_r] &= E\left[\exp\left(\frac{i2dm^2b\pi^2}{(2N+1)^2\Delta t^2}\right)\exp\left(-\frac{i2dr^2b\pi^2}{(2N+1)^2\Delta t^2}\right)\right. \\ &\quad \times \sum_{k=-N}^N n(k)\exp\left(\frac{iak^2\Delta t^2}{2b} - \frac{i2\pi km}{2N+1}\right) \\ &\quad \times \sum_{l=-N}^N x(l)\exp\left(-\frac{ial^2\Delta t^2}{2b} + \frac{i2\pi lr}{2N+1}\right) \\ &= E\left[\exp\left(\frac{i2d(m^2 - r^2)b\pi^2}{(2N+1)^2\Delta t^2}\right)\sum_{k=-N}^N \sum_{l=-N}^N n(k)n(l)\right. \\ &\quad \times \exp\left(\frac{ia(k^2 - l^2)\Delta t^2}{2b} - \frac{i2\pi(km - lr)}{2N+1}\right) \\ &= \exp\left(\frac{i2d(m^2 - r^2)b\pi^2}{(2N+1)^2\Delta t^2}\right)\sum_{k=-N}^N \sum_{l=-N}^N p(k-l) \\ &\quad \times \exp\left(\frac{ia(k^2 - l^2)\Delta t^2}{2b} - \frac{i2\pi(km - lr)}{2N+1}\right) \end{aligned} \quad (15)$$

From (14) and (15), the final result of the covariance between a_m and a_r becomes

$$\begin{aligned} \text{cov}(a_m, a_r) &= E[a_m \bar{a}_r] - E[a_m]E[\bar{a}_r] \\ &= \exp\left(\frac{i2d(m^2 - r^2)b\pi^2}{(2N+1)^2\Delta t^2}\right) \\ &\quad \times \sum_{k=-N}^N \sum_{l=-N}^N (p(k-l) - u_n^2) \\ &\quad \times \exp\left(\frac{ia(k^2 - l^2)\Delta t^2}{2b} - \frac{i2\pi(km - lr)}{2N+1}\right) \end{aligned} \quad (16)$$

In particular, for each $m \in [-N, N]$, the variance of a_m is as follows:

$$\begin{aligned} D[a_m] &= \text{cov}(a_m, a_m) = E[a_m \bar{a}_m] - E[a_m]E[\bar{a}_m] \\ &= \sum_{k=-N}^N \sum_{l=-N}^N (p(k-l) - u_n^2) \\ &\quad \times \exp\left(\frac{ia(k^2 - l^2)\Delta t^2}{2b} - \frac{i2\pi m(k-l)}{2N+1}\right) \end{aligned} \quad (17)$$

This concludes the calculation of covariance matrix. With the mean and covariance matrix of the noise after DLCT, its distribution is completely characterised.

4 Specific cases

4.1 Noise after DFT

When $A = (0 \ 1; -1 \ 0)$, the DLCT becomes DFT. In this subsection, we will discuss the properties about the noise after DFT using the derived results in this paper.

In this case, the mean can be regarded as the DFT of $f(t) = u_n$, so it reduces to

$$\begin{aligned} m=0, \quad \exp\left(-\frac{i2\pi km}{2N+1}\right) &= 1, \quad E[a_m] = (2N+1)u_n \\ m \neq 0, \quad \sum_{k=-N}^N \exp\left(-\frac{i2\pi km}{2N+1}\right) &= 0, \quad E[a_m] = 0 \end{aligned} \quad (18)$$

From (18), it is clear that the mean values of the DFT coefficients are not influenced by the noise, except for the $m=0$ component, it is the same as the classical result on the DFT domain [22].

The covariance matrix of noise after DFT is

$$\begin{aligned} \text{cov}(a_m, a_r) &= E[a_m \bar{a}_r] - E[a_m]E[\bar{a}_r] \\ &= \sum_{k=-N}^N \sum_{l=-N}^N (p(k-l) - u_n^2) \\ &\quad \times \exp\left(-\frac{i2\pi(km - lr)}{2N+1}\right) \end{aligned} \quad (19)$$

When the DLCT reduces to the DFT, from (18) and (19), it is same as the previous work about noise influence on the DFT domain [22]. So the proven theories can be verified through the classical result. Compared with the classical result, the mean value of the DLCT coefficients is partly influenced by the noise with the mean values of the DFT coefficients not influenced by the noise except for the $m=0$ component. It is the main difference with the classical result.

4.2 AWGN after DLCT

Mostly, the noise signal is AWGN [31]. It satisfies our restriction of noise. As a result, we will calculate the covariance matrix of AWGN and finally prove that the AWGN after DLCT is an AWGN in this subsection.

Define $n(k)$ as AWGN sequence. u_n is the mean of $n(k)$, σ_n^2 is the variance of $n(k)$. Then according to the property of AWGN, the correlation function is given by

$$p(m) = \begin{cases} \sigma_n^2, & m=0 \\ 0, & m \neq 0 \end{cases} \quad (20)$$

Let $p_a(m)$ be the correlation function of AWGN after DLCT, from (20) as well as the previous results (16), we obtain the autocorrelation function of AWGN as

$$\begin{aligned}
 E[a_m \bar{a}_r] &= \exp\left(\frac{i2d(m^2 - r^2)b\pi^2}{(2N+1)\Delta t^2}\right) \sum_{k=-N}^N \sum_{l=-N}^N p(k-l) \\
 &\quad \times \exp\left(\frac{ia(k^2 - l^2)\Delta t^2}{2b} - \frac{i2\pi(km - lr)}{2N+1}\right) \\
 &= \exp\left(\frac{i2d(m^2 - r^2)b\pi^2}{(2N+1)\Delta t^2}\right) \\
 &\quad \times \sum_{k=-N}^N \left(\sigma_n^2 \exp\left(-\frac{i2\pi k(m-r)}{2N+1}\right)\right)
 \end{aligned} \quad (21)$$

By (21), it is obvious that

$$E[a_m \bar{a}_r] = \begin{cases} (2N+1)\sigma_n^2, & m = r \\ 0, & m \neq r \end{cases} \quad (22)$$

Let $p_a(m)$ be the correlation function of AWGN after DLCT, the final result is

$$p_a(m) = \begin{cases} (2N+1)\sigma_n^2, & m = 0 \\ 0, & m \neq 0 \end{cases} \quad (23)$$

In the previous proof, the $n(k)$ after DLCT satisfies a Gaussian distribution. Therefore, according to (23) and previous work, the AWGN after performing the DLCT is also AWGN.

5 Simulations

The following examples are given in this section to verify the achieved results above. In the following, we take 1025 samples and assume that the sampling frequency is 1024 Hz. Simulation tests are implemented with MATLAB 2016a.

5.1 Properties of noise after DLCT

First, we discuss the properties of noise in the LCT domain. We choose different kinds of noise and put them in the DLCT algorithm with different parameter matrix.

Fig. 1a is the image with parameter matrix $A = (314 \ 1; -10 \ 0)$ and the white Gaussian noise $n(t) \sim \text{wgn}(0.001, 0.0005^2)$. As we see, the energy of noise on LCT domain gather in the frequency interval $[0, 100]$, which is good agreement between practice and theory. Fig. 1b use the same DLCT with Fig. 1a and coloured noise sequence $n'(k) = n(k) + 0.5n(k-1)$ with $n(k) = n(k\Delta t)$. Compared with Fig. 2a, it demonstrates that the properties of coloured noise in the LCT domain are similar to the properties of the white noise. Fig. 1c is the image which reduces to the FT domain, with the noise same as shown in Fig. 2a. The $m=0$ component nearly equals to $20\log(1025u_n) = 2$ dB, which coincides very well with the practical results. Fig. 1d is also the FT domain with $n(t) \sim \text{WGN}(0.001, 0.00005^2)$. Compared with Fig. 1c, it illustrates that intensity of noise on LCT domain is also related to the variance of noise. It is clear that the simulation results are coinciding with the theorems we derived very well.

5.2 Signal with noise in LCT domain

We then discuss the noise influence on signal in LCT domain. Let define a signal $f(t) = 0.9\exp(100\pi t + 2\pi t^2) + 0.1\exp(400\pi t + 2\pi t^2)$, parameter matrix $A = (-12.4 \ 1; -1 \ 0)$, the noise is WGN, SNR = 60 dB, the variance is 0.0005^2 . Fig. 2a plots the spectrum of this signal without noise in LCT domain and Fig. 2b plots the spectrum of this same signal with WGN in LCT domain. Compared with Fig. 2a, we can find both of them has the amplitude spectrum line at 50 and 200 Hz. However, at the $m=0$ component, Fig. 2b also has value closed to 2 dB. Moreover, the energy of the signal in Fig. 2a is nearly zero beside 50 and 200 Hz components, but there is a little energy on all of the frequency components made by noise, which is closed to 60 dB. It is clear that the simulation results are satisfied with the theorems we derived very well.

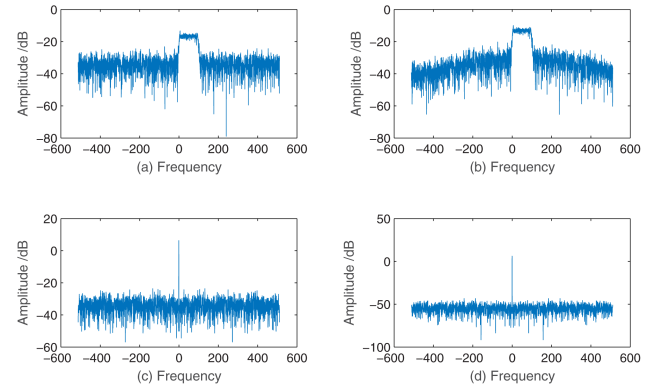


Fig. 1 Properties of noise in LCT domain

(a) Parameter matrix $A = (314 \ 1; -10 \ 0)$, $n(t) \sim \text{WGN}(0.001, 0.0005^2)$, (b) Parameter matrix $A = (314 \ 1; -10 \ 0)$, $n'(k) = n(k) + 0.5n(k-1)$ with $n(k) = n(k\Delta t)$, (c) FT domain, with the noise same as in Fig. 1a, (d) FT domain with $n(t) \sim \text{WGN}(0.001, 0.00005^2)$

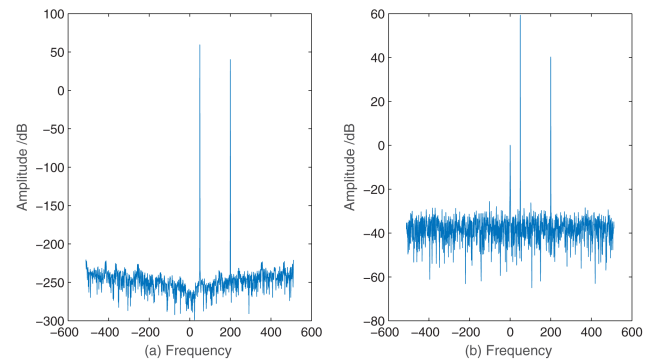


Fig. 2 Signals in LCT domain

(a) The spectrums of no noise signal in LCT domain, (b) The spectrum of WGN noise signal in LCT domain

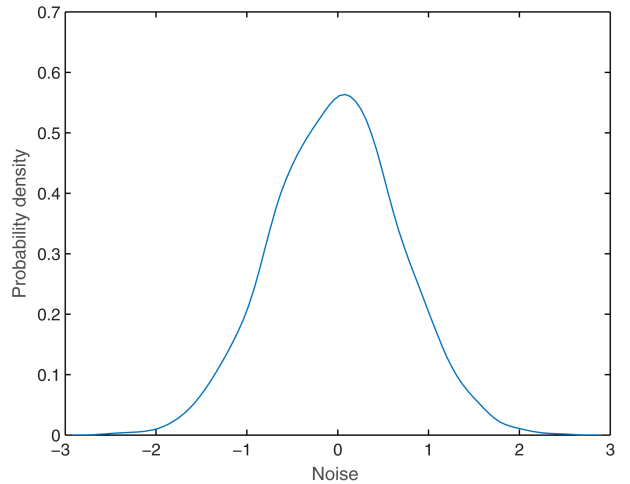


Fig. 3 Probability density function of noise after performing the DLCT

5.3 Noise properties on other DLCT method

Finally, we discuss the noise properties of other DLCT method. For example, Fig. 3 shows the distribution of probability density function of noise after performing the DLCT method in [32]. The number of points is $N=2000$, and the noise sequence is considered to be white. In Fig. 3, the distribution is the 'best fitted' Gaussian distribution. It is easy to see that the simulation results are coinciding with the theorems we derived very well.

According to the simulation, it also shows that the theorems have nothing to do with the kind of the DLCT method, so by the theorems and simulation, we can use other DLCT method to finish the whole proof.

6 Potential applications

The derived theory for noise influence in the DLCT domain states that a noise after DLCT can be estimated with a few assumptions of noise. So the derived theory has many potential applications.

First, the AWGN is still an AWGN after DLCT, it can be widely used in many fields, we can convert a signal with AWGN after DLCT to the signal after DLCT with an AWGN then simplify the signal processing problem. Then the immediate application of the proposed theory can be found in spectral estimation in communication systems constructed by the LCT and limited-angle tomography in imaging reconstruction, where only limited observation data are available [33]. What is more, the DLCT is the discrete version of the continuous LCT, which is very useful for wave propagation analysis and optics. So the derived theory can be also useful for increasing the resolution and/or field of view in an optical imaging system and reconstructing wavefields or beam profiles from partial measurements [34]. Finally, some of the ideas presented in this correspondence may be extended to study some problems in the canonical domain, such as non-uniform sampling and reconstruction, compressed sensing, non-ideal sampling, oversampling, irregular sampling, approximation and wavelet sampling [35].

7 Conclusions

In this paper, we discuss the influence on noise after a DLCT. At first, we show one definition of LCT and DLCT. Then, prove that the additive noise after a DLCT can be considered as having a Gaussian distribution by the central limit theorem and calculate the mean and covariance matrix of it. The mean of the distribution is zero except for the certain interval of components where there is a known bias. What is more, we also prove that the AWGN after DLCT is also an AWGN. Finally, the simulations verify the conclusion we have proven. It is very useful for applications in digital signal processing, it is also important for parameter estimation problems where the knowledge of the noise on the measurements is very valuable.

8 Acknowledgments

This work is supported by the National Natural Science Foundation of China (no. 61671063), and also by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China (no. 61421001).

9 References

- [1] Moshinsky, M., Quesne, C.: 'Linear canonical transformations and their unitary representations', *J. Math. Phys.*, 1971, **12**, (8), pp. 1772–1783
- [2] Ozaktas, H.M., Zalevsky, Z., Kutay, M.A.: 'The fractional Fourier transform with applications in optics and signal processing' (Wiley, New York, 2001)
- [3] Zhang, F., Tao, R., Wang, Y.: 'Discrete linear canonical transform computation by adaptive method', *Opt. Express*, 2013, **21**, (15), pp. 18138–18151
- [4] Zhao, J., Tao, R., Wang, Y.: 'Sampling rate conversion for linear canonical transform', *Signal Process.*, 2008, **88**, (11), pp. 2825–2832
- [5] Tao, R., Li, B.Z., Wang, Y.: 'Spectral analysis and reconstruction for periodic nonuniformly sampled signals in fractional Fourier domain', *IEEE Trans. Signal Process.*, 2007, **55**, (7), pp. 3541–3547
- [6] James, D.F.V., Agarwal, G.S.: 'The generalized Fresnel transform and its applications to optics', *Opt. Commun.*, 1996, **126**, (4), pp. 207–212
- [7] Li, B.Z., Tao, R., Wang, Y.: 'New sampling formulae related to linear canonical transform', *Signal Process.*, 2007, **87**, (5), pp. 983–990
- [8] Deng, B., Tao, R., Wang, Y.: 'Convolution theorems for the linear canonical transform and their applications', *Sci. China F. Inf. Sci.*, 2006, **49**, (5), pp. 592–603
- [9] Onural, L., Gotchev, A., Ozaktas, H.M., *et al.*: 'A survey of signal processing problems and tools in holographic three-dimensional television', *IEEE Trans. Circuits Syst. Video Technol.*, 2007, **17**, (11), pp. 1631–1646
- [10] Wei, D.Y., Li, Y.M.: 'Reconstruction of multidimensional bandlimited signals from multichannel samples in linear canonical transform domain', *IET Signal Process.*, 2014, **8**, (6), pp. 647–657
- [11] Yan, J.P., Li, B.Z., Chen, Y.H., *et al.*: 'Distribution moments associated with the linear canonical transform', *Int. J. Electron.*, 2013, **100**, (4), pp. 473–481
- [12] Cao, Y.J., Li, B.Z., Chen, Y.H.: 'Logarithmic uncertainty relations for odd or even signals associate with Wigner–Ville distribution', *Circuits Syst. Signal Process.*, 2016, **35**, (7), pp. 2471–2486
- [13] Feng, Q., Li, B.Z.: 'Convolution and correlation theorems for the two-dimensional linear canonical transform and its applications', *IET Signal Process.*, 2016, **10**, (2), pp. 125–132
- [14] Guo, Y., Li, B.Z.: 'Blind image watermarking method based on linear canonical wavelet transform and QR decomposition', *IET Image Process.*, 2016, **10**, (10), pp. 773–786
- [15] Pei, S.C., Ding, J.J.: 'Closed-form discrete fractional and affine Fourier transforms', *IEEE Trans. Signal Process.*, 2000, **48**, (5), pp. 1338–1353
- [16] Li, C.P., Li, B.Z., Xu, T.Z.: 'Approximating bandlimited signals associated with the LCT domain from nonuniform samples at unknown locations', *Signal Process.*, 2012, **92**, (7), pp. 1658–1664
- [17] Healy, J.J., Sheridan, J.T.: 'Sampling and discretization of the linear canonical transform', *Signal Process.*, 2009, **89**, (4), pp. 641–648
- [18] Koc, A., Ozaktas, H.M., Candan, C., *et al.*: 'Digital computation of linear canonical transforms', *IEEE Trans. Signal Process.*, 2008, **56**, (6), pp. 2383–2394
- [19] Hennelly, B.M., Sheridan, J.T.: 'Fast numerical algorithm for the linear canonical transform', *J. Opt. Soc. Am. A*, 2005, **22**, (5), pp. 928–937
- [20] Zhao, L., Healy, J.J., Sheridan, J.T.: 'Unitary discrete linear canonical transform: analysis and application', *Appl. Opt.*, 2013, **52**, (7), pp. 30–36
- [21] Healy, J.J., Sheridan, J.T.: 'Fast linear canonical transforms', *J. Opt. Soc. Am.*, 2010, **27**, (1), pp. 21–30
- [22] Schoukens, J., Renneboog, J.: 'Modeling the noise influence on the Fourier coefficients after a discrete Fourier transform', *IEEE Trans. Instrum. Meas.*, 1986, **35**, (3), pp. 278–286
- [23] Schoukens, J., Vandersteen, G., Guillaume, P., *et al.*: 'Frequency-domain system identification using non-parametric noise models estimated from a small number of data sets', *Automatica*, 1997, **33**, (6), pp. 1073–1086
- [24] Kalashnikov, N.A., Challis, R.E.: 'Errors and uncertainties in the measurement of ultrasonic wave attenuation and phase velocity', *IEEE Trans. Ultrason. Ferroelectr. Freq. Control*, 2005, **52**, (10), pp. 1754–1768
- [25] Hwang, J.K., Liu, Y.: 'Noise analysis of power system frequency estimated from angle difference of discrete Fourier transform coefficient', *IEEE Trans. Power Deliv.*, 2014, **29**, (4), pp. 1533–1541
- [26] Sun, Y.X., Zhuang, C.Q., Xiong, Z.H.: 'A scale factor-based interpolated DFT for chatter frequency estimation', *IEEE Trans. Instrum. Meas.*, 2015, **P1**, pp. 2666–2678
- [27] Pandey, N., Hennelly, B.: 'Quantization noise and its reduction in lensless Fourier digital holography', *Appl. Opt.*, 2011, **50**, (7), pp. 58–70
- [28] Xu, T.Z., Li, B.Z.: 'Linear canonical transform and its applications' (Science Press, Beijing, China, 2013)
- [29] Zhao, M., Wu, W.S.: 'Analysis of linear frequency modulated signal', *Aero Weaponry*, 2009, **1**, pp. 25–29
- [30] Besson, O., Ghogho, N., Swami, A.: 'Parameter estimation for random amplitude chirp signals', *IEEE Trans. Signal Process.*, 1999, **47**, (12), pp. 3208–3219
- [31] Wu, Z.H., Huang, N.E.: 'A study of the characteristics of white noise using the empirical mode decomposition method', *Proc. R. Soc.*, 2004, **460**, (2046), pp. 1597–1611
- [32] Pei, S.C., Huang, S.G.: 'Fast discrete linear canonical transform based on CM-CC-CM decomposition and FFT', *IEEE Trans. Signal Process.*, 2016, **64**, (4), pp. 855–866
- [33] Shi, J., Liu, X.P., He, L., *et al.*: 'Sampling and reconstruction in arbitrary measurement and approximation spaces associated with linear canonical transform', *IEEE Trans. Signal Process.*, 2016, **64**, (24), pp. 6379–6391
- [34] Shi, J., Sha, X.J., Zhang, Q.Y., *et al.*: 'Extrapolation of bandlimited signals in linear canonical transform domain', *IEEE Trans. Signal Process.*, 2012, **60**, (3), pp. 1502–1508
- [35] Shi, J., Liu, X.P., Sha, X.J., *et al.*: 'Sampling and reconstruction of signals in function spaces associated with linear canonical transform', *IEEE Trans. on Signal Process.*, 2012, **60**, (11), pp. 6041–6047